

E-Commerce Database Analysis

A comprehensive examination of sales and profitability metrics from our online retail platform. This analysis provides critical insights into monthly performance, product category dynamics, sub-category performance, and customer segment profitability to guide strategic decision-making.

Monthly Sales Performance

Key Findings

The analysis reveals significant seasonal variation in monthly sales performance. November emerged as the peak sales month, likely driven by holiday shopping patterns and promotional campaigns. February showed the lowest sales, potentially reflecting post-holiday slowdown.

This 5.9-fold difference between highest and lowest months highlights the importance of seasonal planning and inventory management strategies.

\$352K

Highest Sales

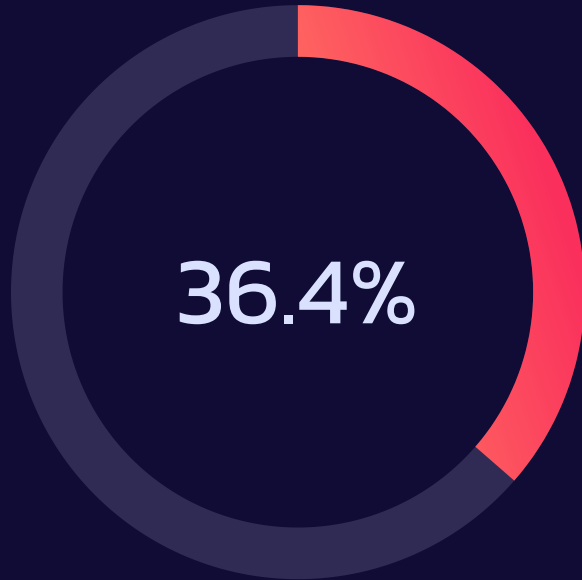
November (Month 11)

\$59.8K

Lowest Sales

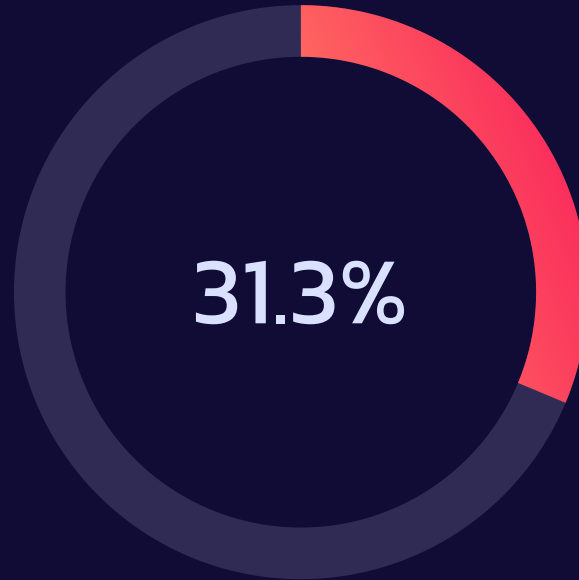
February (Month 2)

Product Category Sales Analysis



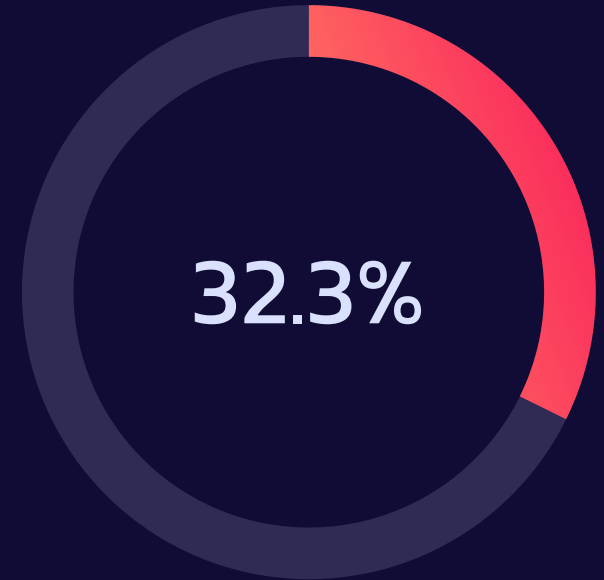
Technology

\$836,154.03



Office Supplies

\$719,047.03



Furniture

\$742,327.02

The category analysis shows Technology leading in sales with \$836,154.03 (36.4% of total), followed by Furniture at \$742,327.02 (32.3%), and Office Supplies at \$719,047.03 (31.3%). The relatively balanced distribution suggests no single category dominates, providing portfolio diversification benefits.

Sub-Category Performance Breakdown

Performance Extremes

At the sub-category level, performance varies dramatically. Phones dominate with \$330,007.05 in sales, representing strong consumer demand for mobile technology. Conversely, Fasteners show minimal sales at only \$3,024.28, likely serving as a niche product with limited appeal.

This 109-fold difference between top and bottom sub-categories suggests opportunities to optimize product assortment by reducing low-performing SKUs and investing in high-performing categories.

\$330K

Phones

Highest

\$3.02K

Fasteners

Lowest

Monthly Profit Analysis

December – Highest Profit

\$43,369.19

Month 12

January – Lowest Profit

\$9,134.45

Month 1

Profit analysis reveals December as the most profitable month with \$43,369.19, coinciding with peak holiday sales. January showed the lowest profit at \$9,134.45, despite decent sales volume, suggesting higher promotional costs or margin compression during post-holiday clearance periods. The 4.7-fold difference in monthly profit highlights the need for strategic pricing and cost management throughout the year.

Profit by Category & Sub-Category

Category-Level Analysis

Technology: \$145,454.95 (50.8%)

Highest profit category, driven by high-margin electronics and accessories

Furniture: \$18,451.27 (6.44%)

Lowest profit category, likely due to shipping costs and lower margins

Sub-Category-Level Analysis

Highest Profit

Copiers: \$55,617.82

High-margin business equipment with strong customer demand

Lowest Profit

Tables: -\$17,725.48

Actual loss, indicating pricing or cost structure issues

Customer Segment Performance



Consumer

Top segment for both sales volume and total profit



Corporate

Middle segment with steady business purchases



Home Office

Lowest segment for both sales and profit metrics

The consumer segment leads in both sales volume and total profit generation, reflecting the broad-based retail nature of the e-commerce platform. Home Office represents the smallest segment, suggesting either limited market penetration or lower purchase frequency. Strategic initiatives should focus on growing the Home Office segment through targeted marketing and product bundling.

Sales to Profit Ratio Analysis

By Customer Segment



Home Office: 7.13

Best profitability ratio

Corporate: 7.68

Strong performance

Consumer: 8.66

Lower efficiency

By Product Category



Technology: 5.75

Best category profitability

Office Supplies: 5.87

Strong margins

Furniture: 40.21

Very low profitability

The sales to profit ratio reveals that lower ratios indicate better profitability. Technology shows the best category performance at 5.75, while Furniture's 40.21 ratio signals serious margin issues requiring immediate attention.

Sub-Category Profitability Insights

Best Profitability

- **Envelopes: 2.37**
Excellent margin performance
- **Labels: 2.25**
Strong profitability
- **Paper: 2.30**
High efficiency

Worst Profitability

- **Machines: 55.91**
Very high ratio
- **Bookcases: -33.08**
Negative profitability
- **Tables: -11.68**
Operating at loss

The sub-category analysis reveals dramatic profitability differences. Low-ratio items like Envelopes, Labels, and Paper demonstrate excellent margin performance. Conversely, negative ratios for Bookcases and Tables indicate actual losses, while Machines shows extremely poor efficiency. Immediate action should focus on addressing loss-making sub-categories through pricing adjustments, cost reduction, or product line rationalization.